

Midair Haptic-Optic Display with Multi-Tactile Texture based on Presenting Vibration and Pressure Sensation by Ultrasound

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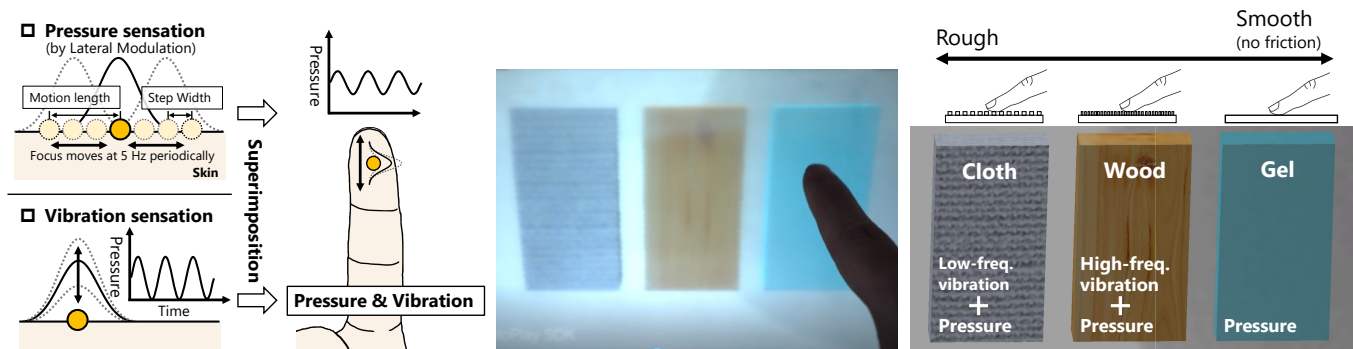


Figure 1: Left) Presentation of pressure sensation and vibration sensation using ultrasound. Center) Presented aerial 3D images with different roughness textures: cloth, wooden board, and gel. Right) Digital data of the presented images.

ABSTRACT

Reproducing the tactile texture of a real object allows humans to interact in virtual reality space with a high sense of immersion. In this study, we develop a midair haptic-optic display with multi-tactile texture using focused ultrasound. In this system, users can touch aerial 3D images with a realistic texture without wearing any devices. For rendering realistic texture, our system simultaneously presents vibration and static pressure sensation. The presentable sensation of the previous ultrasound system is limited to vibrations. The combination of the sensations can improve the reality of rendered tactile texture by ultrasound. In the demo, we presented three textures with different roughness: cloth, wooden board, and gel.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

Multi-texture, haptic display, midair haptics, focused ultrasound

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1 INTRODUCTION

When touching a physical object, humans perceive the tactile texture of the surface. Rendering tactile texture allows users to touch an object in virtual reality space with a high sense of immersion.

In midair ultrasonic haptics, rendering texture is a challenging problem since the presented pressure of ultrasound is weak. An airborne ultrasound tactile display (AUTD) is a device for midair haptics in which ultrasound transducers are arranged in an array [Carter et al. 2013; Hoshi et al. 2010]. An AUTD can present various pressure distributions to human skin in a noncontact manner. With AUTDs, users can feel tactile stimuli without wearing any device. However, since the presented force by an AUTD is weak (about a few grams), spatiotemporal modulation of the presented pressure distribution is always necessary to increase its perceived intensity [Takahashi et al. 2019]. The modulation has limited the presentable sensation of an AUTD to vibration thus, rendering realistic texture by an AUTD has been difficult.

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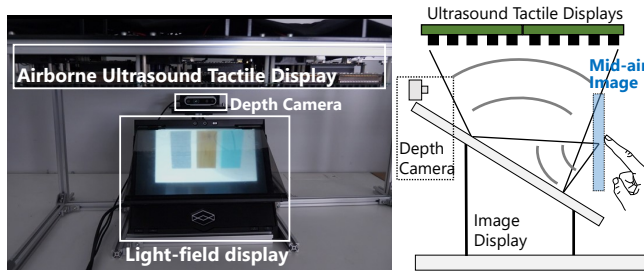


Figure 2: System configuration

Toward rendering texture, we recently found the modulation condition in which an AUTD presents a strong static pressure sensation (Fig. 1-Left) [Morisaki et al. 2020]. A pressure sensation is essential for reproducing a texture since humans always perceive it when touching objects and one of the tactile receptors perceiving texture responds to pressure [Bolanowski Jr et al. 1988]. In AUTDs, the pressure sensation can be presented by Lateral Modulation (LM), which was originally proposed to present a vibration sensation [Takahashi et al. 2019]. In LM, an ultrasound stimulus point (focus) periodically moved along the skin surface with constant pressure. When the moving frequency is low (5 Hz) and the moving step width is fine (0.2 mm), a strong pressure sensation is evoked.

In this study, for rendering texture in midair, we propose to simultaneously present vibration and pressure sensation using AUTDs. Based on the rendering method, we develop a midair haptic-optic display that allows participants to actively touch aerial 3D images with realistic tactile textures. Previous studies added haptic feedback to aerial images using AUTDs, but they focused on reproducing a shape and not rendering textures [Makino et al. 2016; Monnai et al. 2014]. Sakiyama et al. measured the pressure distribution of a real object moving on the sensor surface and reproduced it on a stationary palm using AUTDs [Sakiyama et al. 2020]. However, their system has no interactivity and the presentable sensation is limited to vibrations or some dynamically changing feelings. Our display is the first interactive system adding not only vibration but also pressure sensation to aerial images using ultrasound.

2 MIDAIR HAPTIC-OPTIC DISPLAY

The developed haptic-optic display (Fig. 2) consists of six AUTDs presenting tactile stimulus, a light field display (Holo Player One, Looking Glass Factory), and a depth camera (Intel Real Sense D435) for sensing the position of a finger. One AUTD unit we used has 249 ultrasound transducers driving at 40 kHz. When a finger contacts the presented image, the AUTDs present ultrasound stimulus to the contact position.

2.1 Presentation Method of Vibration and Static Pressure Sensation

This system simultaneously presents vibration and static pressure sensation using ultrasound to render realistic textures (Fig. 1-left). A pressure sensation is presented by an LM stimulus at 5 Hz. The spatial step width of the focus movement is 0.2 mm and the movement width is 3 mm with reference to previous experiment [Morisaki

et al. 2020]. The moving direction of the focus is in the direction along with a finger.

Vibration sensation is presented by modulating the pressure at the moving focus in the LM. The ratio of the vibration sensation to the pressure sensation in the presented texture is controlled by the amplitude of the superimposed vibration waveform. In all stimuli, the maximum presentation force is fixed to maintain high perceptual intensity.

2.2 Rendering Surface Texture Sensation

In this demo, we present three textures with different roughness (Fig. 1-center and right), which is the fundamental sensation in tactile space. We chose cloth as the roughest texture, wooden board as the fine roughness texture, and gel as the smoothest texture with no friction. The presentation stimulus is adjusted to match each image with reference to the spatial wavelength of each surface. We adapted sinusoidal wave vibration of 50 Hz for cloth and 200 Hz for wood. For gel, only pressure sensation is presented. We subjectively confirmed that the reality of the texture was improved by simultaneously presenting the vibration and the pressure sensation compared with a vibration-only case.

3 CONCLUSION

In this study, for rendering a tactile texture in midair, we proposed to simultaneously present a static pressure sensation and a vibration sensation using focused ultrasound. Based on the rendering method, we developed a midair haptic-optic display that allows participants to touch aerial 3D images with realistic surface texture. We believed that the combination sensation improved the reality of the texture.

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